DialyLight:
A Blue Light-Based Antimicrobial Device for Hemodialysis Catheter Disinfection


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Problem Definition

- Annually, more than 800k people are diagnosed with End-Stage Renal Disease (ESRD)
- ~70% require dialysis treatment

81% of dialysis patients rely on Hemodialysis Catheters (HC)

HC patients experience sepsis at a prevalence rate of 41.5%

Need Statement

There is a need for a device that safely and effectively ***eliminates bacterial buildup*** at the catheter exit site in ***early-stage HemoDialysis (HD) patients.***

Market Analysis

- **Total Addressable Market (TAM):** New ESRD patients undergoing HD
- **Serviceable Available Market (SAM):** Patients undergoing catheter-based HD
- **Serviceable Obtainable Market (SOM):** Patients willing to switch to our device
- **Estimated product cost:** $2000

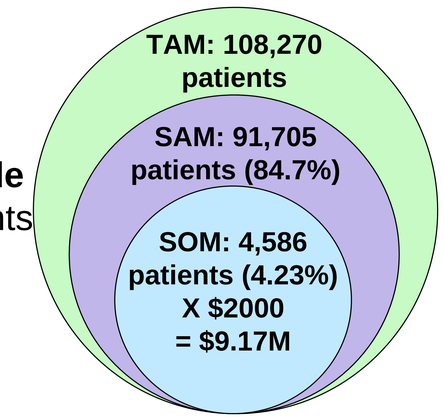


Figure 1. Market Analysis for DialyLight

Functional Requirements

Power/Wavelength - Emit a 450-470 nm λ and a power density of 50-90 mW/cm²

Non-Clinical Application - Permit use in a home/non-clinical setting

Attachment - Secure device to port site and surrounding skin

Engineering Design

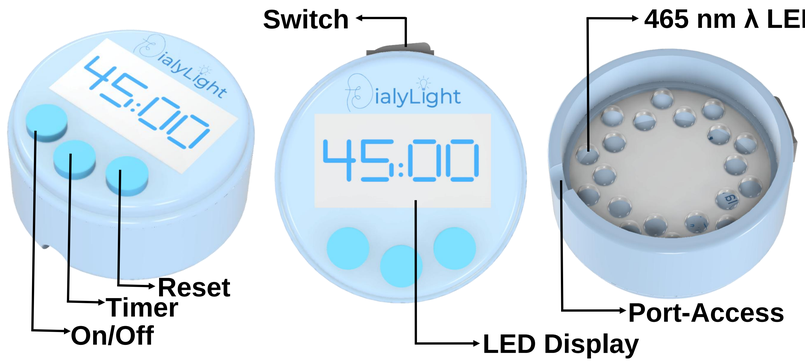


Figure 2. Figure 2. Schematic diagram of the blue-light based catheter disinfection device.

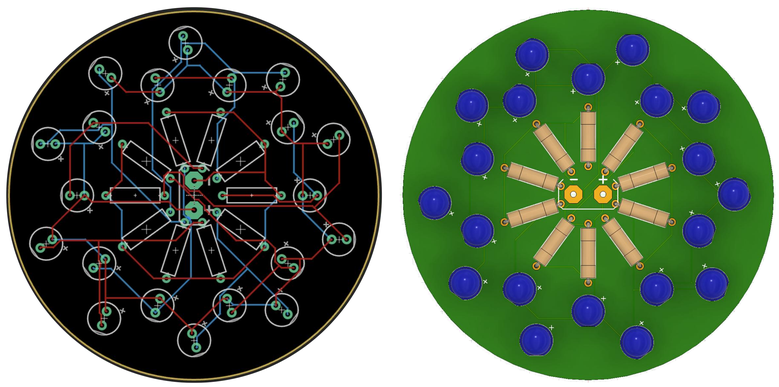


Figure 3. 2D PCB with Traces (left) and 3D PCB (right) as utilized in our final device set-up.

Engineering Design *(continued)*

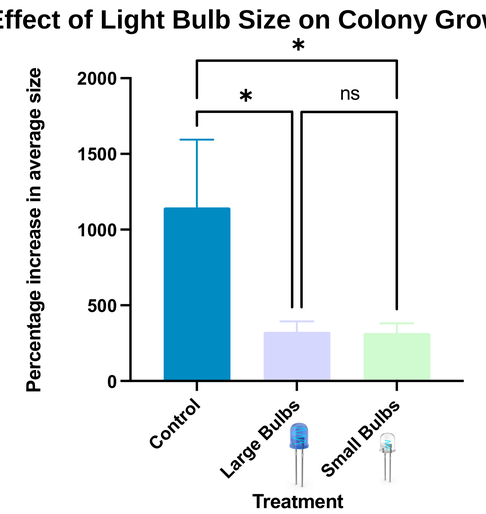


Figure 4. Percentage increase in average bacterial colony size 24 hours post-treatment. * p<0.05

- Percent increase in average bacterial colony size was calculated from triplicate experiments
- One-way ANOVA with Tukey HSD post-hoc test revealed significant differences between bulb treatments and control (**Figure 4**) → Blue light suppresses bacterial growth
- No significant difference between large and small bulb treatments

Conclusions & Future Directions

DialyLight (450-470 nm) targets bacterial buildup, is cost-effective and applicable in non-clinical settings.

Future Directions:

- Clinical trials → Assessing real world efficacy
- Design optimization for scalability, comfort, and affordability
- Application expansion to other catheter/open port site procedures

References: 1. CDC (2024), 2. FDA (2025), 3. NIDDK USRDS (2024), 4. NKF (2023), 5. Pires et al. (2013), 6. SparkFun (2025)

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